

Within-system discrimination does not imply cross-system transfer: interaction foundation models in structure-based generative design

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Abstract

Pretrained molecular-interaction foundation models are increasingly adapted for structure-based generative design under the assumption that their ability to recognise native-like binding geometries will transfer to guiding generative processes in novel pockets. In this work, we demonstrate a sharp regime boundary in the transferability of these models. We find that a foundation model can discriminate interaction geometry almost perfectly *within* a single complex, yet fails to transfer its predictive power *between* complexes. Furthermore, we identify a mechanistic dose-response consequence: optimising the model’s interface distance as a training-time critic does not weakly assist pocket fit but actively *trades against* it, provably lowering the objective while generated molecules dock worse and reproduce fewer reference contacts. Finally, we show that standard evaluations for structure-based generative models cannot resolve the effects they are frequently used to claim. By pairing every negative result with a control that proves the measurement could have detected the positive, we quantify these detection limits, recommending that future generative evaluations report a measured detection limit and a same-scale null alongside point estimates.

1 Introduction

Foundation models for molecular interactions are widely assumed to transfer seamlessly to downstream tasks in structure-based drug design (SBDD). As these models are exposed to broad swaths of biological data, their encoded representations are anticipated to generalise out-of-distribution, particularly when discriminating active from inactive compounds, or when guiding generative processes toward physically viable interactions.

In this work, we present a sharp regime boundary governing the utility of these representations. We find that a pretrained molecular-interaction foundation model (ATOMICA (Fang et al. 2025)) discriminates interaction geometry almost perfectly *within* a single protein-ligand complex. However, it fails to transfer this capability *between* distinct complexes. Applications involving these models divide cleanly along this line, and the boundary robustly predicts which applications will succeed.

Furthermore, we identify the mechanistic dose-response consequences of this boundary. Optimising the model’s interface distance as a training-time critic does not weakly help pocket fit; rather, it *trades against* it. We show that the critic objective provably falls while the generated molecules dock worse and reproduce fewer reference contacts. Lastly, we establish that standard evaluation metrics in SBDD cannot resolve the effects they are routinely used to claim, quantifying by how much these metrics fall short of necessary detection limits.

Crucially, every negative result reported here is paired with a specific control proving that the measurement *could* have detected the positive effect, transforming these findings from an absence of evidence into concrete

evidence of absence. The remainder of this paper is structured as follows. Section 2 outlines the ATOMICA representations, the diffusion generator, the benchmark setups, and the controls employed. Section 3 details the evidence supporting our three main claims. Section 4 discusses the implications of the regime boundary, the mechanisms underlying the transfer failure, and provides actionable recommendations for evaluating structure-based generative models.

2 Methods

We study the ATOMICA model (Fang et al. 2025) applied as a pretrained interaction foundation model. Our analysis evaluates its representations across two primary segments: a two-segment interface encoding and the PS_300 fragmentation scheme, contrasting graph versus unit representations.

To robustly evaluate the capabilities of the model, we use a comprehensive pose benchmark and a diffusion generator equipped with a critic objective. Our diffusion model builds on recent equivariant diffusion architectures for SBDD (Schneuing et al. 2024). We assess performance using established structural evaluation frameworks. Notably, we employ the interaction fingerprint metrics from PoseCheck (Harris et al. 2023), which evaluates the biological realism of generated poses, and use their qualitative finding that generated molecules often make fewer key interactions than reference ligands as independent corroboration of our null and ceiling expectations. We also consider binding-specificity metrics akin to Delta Score (Gao et al. 2024), physical validity checks analogous to PoseBusters (Buttenschoen et al. 2024), and established benchmarks such as GenBench3D (Baillif 2024), CBGBench (Li et al. 2025), and CrossDocked2020 (Francoeur et al. 2020).

When assessing interaction hotspot fields, we deploy the validation protocol of Radoux et al. (2016) (Radoux et al. 2016). Docking validations are performed using smina (Koes et al. 2013) and AutoDock Vina (Trott and Olson 2010), and molecular manipulations rely on RDKit (RDKit 2024). We explicitly eschew learning supervised pharmacophore readouts directly from probe representations (as done in PharmacoNet (Seo and Kim 2024)) to isolate the intrinsic, training-free signal of the foundation model.

Controls. The core of our methodology is the rigorous pairing of every metric with a same-scale null. We rely on the following controls: - **Permuted-weight control:** Evaluates architectures without learned chemical information. - **Buriedness control:** Distinguishes protein neighbour-count confounds from genuine interaction field signals. - **Placement null:** Provides a baseline of randomly positioned but structurally sound compounds. - **True-binder ceiling:** Establishes the maximum expected signal from a known, distinct binder. - **Matched decoys & seed-only noise floors:** Quantify the variance strictly attributable to sampling and data splits rather than physical affinity.

Gotcha: During initial tests, a naive `pocket_pool` representation scored the best raw AUROC of any variant (0.949). However, testing it with random weights yielded 0.923. The high performance was an artefact of architecture and geometry, not learned chemistry. This demonstrates why every representation claim in this study is strictly paired with a same-architecture, same-scale, no-learned-information comparison.

Pre-registration. Analysis plans for these experiments were committed before the results existed (specifically, `results/specificity/ANALYSIS_PLAN.md` and `results/interface_fidelity/ANALYSIS_PLAN.md`), guaranteeing that our evaluation criteria were not post-hoc adapted to the findings.

3 Results

3.1 Within-system discrimination is near-perfect

We first test the foundation model’s capacity to discriminate poses within a single complex. We use a clash- and composition-controlled pose benchmark with 30 poses per class. As shown in Table 1, the linear probe on the representation achieves an AUROC of 1.000, establishing an *upper bound on extractable signal* that is fitted on the system it scores; it is not a competing number but rather a ceiling of theoretical detectability. Notably, the rotation head is anti-correlated: displaced poses score *lower* rotational correction than native ones.

Table 1: Within-system discrimination performance on the clash- and composition-controlled pose benchmark. The 1.000 AUROC represents an upper bound on extractable signal.

scorer	AUROC	Spearman vs RMSD	needs fitting
min contact distance (trivial baseline)	0.727	—	no
training-free denoising energy	0.787	+0.476	no
linear probe on the representation	1.000	+0.927	yes, per system

We independently confirm this on a larger set of 92 targets and 1,662 poses. Because every pose of a target is the same molecule rigidly displaced, composition is controlled by construction. The pretrained `graph_cosine` distance achieves an AUROC of 0.926, vastly outperforming the permuted-weight control (0.697) (Table 2).

Table 2: Independent confirmation on a larger set (92 targets, 1,662 poses). Pretraining provides the critical signal, leaving the permuted-weight control near the baseline in the low-RMSD regime.

metric	$\rho(\text{all})$	$\rho(<4 \text{ \AA})$	AUROC
<code>graph_cosine</code> (pretrained)	+0.386	+0.558	0.926
<code>graph_cosine</code> (permuted weights)	+0.149	+0.061	0.697
contacts (no-learning floor)	+0.253	+0.355	0.837
smina (reference)	+0.281	+0.465	0.844

3.2 Cross-system transfer fails

We next evaluated the representations on a 100-target CASF-style benchmark (Su et al. 2019) built from open RCSB data, consisting of 72 solvable targets and 1,674 poses. Evaluation was performed out-of-fold under GroupKFold by target, and the docking power metric measured whether the top-ranked pose fell within 2 Å of the crystal structure.

Every evaluated variant sits well above the 15.7% random floor, indicating that the head learns something real and transferable (Table 3). However, it is fundamentally uncompetitive with smina, a fast, free, established baseline (59.7%).

Table 3: Cross-system transfer evaluation on a 72-target benchmark. Transferable signal exists (above the random floor) but fails to outcompete standard baselines like smina.

scorer	docking power	hits	mean per-target	
			Spearman	vs smina
random floor	15.7%	—	—	—
smina	59.7%	43/72	—	—
graph (32-d)	55.6%	40/72	+0.371 ± 0.084	p = 0.70
pocket_pool (96-d)	41.7%	30/72	+0.344 ± 0.077	p = 0.019, worse
all-block (288-d)	63.9%	46/72	+0.400 ± 0.074	p = 0.65

We must report a retraction in the main text regarding these results. An earlier 22-target preliminary assessment had pocket_pool at 68.2% with a paired Spearman gain of +0.127 ($p \approx 0.07$). At 72 targets, this effect reversed entirely, measuring 41.7% and -0.027 ($p = 0.52$). The reversal was driven by selection bias (six feature sets were evaluated, and the best was initially reported) and sample size (22 targets could not resolve differences, with every McNemar $p \geq 0.69$). Recording a selection bias does not remove it, making rigorous out-of-fold validation and pre-registration essential.

3.3 Interaction hotspot fields carry no signal

We probed interaction hotspot fields using chemical probes on a grid, scored with the training-free denoising energy under the validation protocol of Radoux et al. (2016) (Radoux et al. 2016). For CDK2 (1H1S chain A) over a 5 Å site, we tested 1,683 non-clashing points.

As shown in Table 4, the median percentile for the matched probe is 52.4. Without a control, landing in the 98th percentile might seem successful; however, our buriedness control reaches 98.2. Protein neighbour count alone beats the interaction field. Consequently, the type specificity remains near chance (0.107 vs 0.167), proving the field carries no detectable chemical signal.

Table 4: Interaction hotspot field signal evaluation. The buriedness control outperforms the field, exposing protein neighbour count as the true driver of percentile rank.

measure	value	reference
median percentile, matched probe	52.4	Radoux: 97 (fragments), 72 (leads)
median percentile, buriedness control	98.2	the confound
median percentile, random placement	52.2	the floor
type specificity	0.107	chance = 0.167

3.4 Pocket-only encodings are degenerate

We evaluated 99 pockets differing exclusively in block vocabulary to understand the behaviour of pocket-only encodings (Table 5). A mean pairwise cosine similarity of 1.0000 across different pockets indicates that every pocket mapped to the exact same direction. Conditioning on pocket identity was not merely attenuated; it was absent. Fixing the vocabulary to a per-residue definition barely altered this (0.9917). Only by supplying the second segment (the ligand) does the cosine fall to 0.9248 over distinct targets. Therefore, the binding defect originates from the missing interaction partner, not the block vocabulary.

Table 5: Pocket-only encoding degeneracy. High cosine similarities indicate identical directional mapping for all pockets when the ligand partner is omitted.

	old [GLB, UNK]	per-residue
blocks per pocket	2.0	56.3 (17.9 distinct types)
mean pairwise cosine between different pockets (graph)	1.0000	0.9917
mean pairwise cosine (unit)	1.0000	0.9999
composition probe R^2 (graph)	0.201	0.176

This degeneracy explains an earlier observation across 100 pockets where valid molecule metrics improved (QED $0.424 \rightarrow 0.483$; diversity $0.731 \rightarrow 0.684$) with no target-aware gain: the model behaved as a generic drug-likeness prior. This is a worked example of a target-independent metric being mistakenly interpreted as evidence of pocket fit.

3.5 The critic: the objective moves, the molecules do not

We integrated the distance metric as a critic objective in the diffusion generator: $L = L_{\text{diffusion}} + \lambda(t) \cdot d(\text{ATOMICA}(\text{pocket}, \hat{x}_0), \text{ATOMICA}(\text{pocket}, x_{\text{true}}))$. The ATOMICA model was frozen, with LoRA applied

to the EGNN. Crucially, the objective calibrates λ by gradient norm, not loss ratio; achieving a 10% gradient share requires $\lambda = 53.6$ through the input-side adapter but only 0.688 through LoRA.

At $\lambda = 0.7$, the critic does not move its own objective. A paired replay across 475 samples with shared seeds produced differences that flipped signs across seeds (Table 6), proving previous reports of a -37.9% logged metric fall were seed artefacts.

Table 6: Paired replay at $\lambda = 0.7$ across shared seeds. The sign flips across seeds, demonstrating that the apparent fall in the logged metric was a seed artefact.

seed	critic	control	Δ	critic lower at	p
r0	0.002972	0.003168	-0.000197	261/475	0.006
r1	0.003109	0.002976	+0.000133	223/475	0.030
r2	0.003067	0.003112	-0.000045	243/475	0.849

At $\lambda = 20$, the critic actively degrades performance. While its own ATOMICA objective improves ($p = 7e-7$), every downstream readout is worse or flat (Table 7). The molecules dock 0.232 kcal/mol worse into their own pockets.

Table 7: Downstream readouts at $\lambda = 20$. The critic successfully optimises its internal objective but consistently degrades physical pocket fit.

readout	$\lambda = 20$ vs control
its own ATOMICA objective	improves , 2.8 control-seed sd, $p = 7e-7$, paired
diffusion loss	+0.0078 (~ 7 sd of the seed spread)
docking into its own pocket	0.232 kcal/mol worse , $p = 0.001$
pocket specificity	-0.055 ± 0.047 , 17/44 pockets, $p = 0.21$ (MDE 0.131)
interface fidelity	no gain; contact recovery slightly worse

3.6 Interface fidelity, and the dose-response

To assess interface fidelity, we deployed two co-primary fingerprints against the reference ligand across 44 pockets: FP-A (residue-contact Tanimoto, 4.0 Å, heavy atoms) and FP-B (ProLIF interaction fingerprint (Bouysset and Fiorucci 2021), recall), matching constructs used by PoseCheck (Harris et al. 2023).

First, we establish the measurable scale in Table 8. The entire span under study equates to approximately 0.6 Å of rigid displacement.

Table 8: Scale of interface fidelity fingerprints before comparison. The usable range is narrow, emphasising the need for precise controls.

	FP-A Tanimoto	FP-A recall	FP-B Tanimoto	FP-B recall
placement null	0.5107	0.6261	0.3115	0.4696
real arms, pooled	0.6277	0.7786	0.3764	0.5769
true-binder ceiling	0.7596	0.8333	0.5251	0.7500
usable range	0.2489	0.2073	0.2136	0.2804

Comparing the critic arm (`critic_lambda20_r0`) against the control, we find no fidelity gain, and suggestive evidence of active degradation (Table 9). At $\lambda = 0.7$, the cross-arm difference is negligible (-0.00009 on FP-A). At $\lambda = 20$, the shift clears seed noise by large margins, proving the dose-response is readable. We conclude there is no fidelity gain, though the active degradation rests on a single seed and remains suggestive.

Table 9: Interface fidelity comparison at $\lambda = 20$. The critic provides no fidelity gain, and actively degrades the FP-A Tanimoto score.

statistic	Δ	% of range	improved	p	s_arm
FP-A Tanimoto, mean	-0.00800 ± 0.00214	-3.22%	12/44	0.0002	0.00756
FP-A Tanimoto, top-3	-0.00738 ± 0.00406	-2.97%	17/44	0.134	0.01011
FP-B recall, mean	$+0.00237 \pm 0.00428$	$+0.84\%$	20/44	0.690	0.00538
FP-B recall, top-3	$+0.00258 \pm 0.00415$	$+0.92\%$	10/44	0.639	0.00827

3.7 Detection limits

Our final analysis quantifies the detection limits of standard evaluations: - **Cross-docking specificity** (Gao et al. 2024) reveals a Minimum Detectable Effect (MDE) of 0.312 before matched decoys, and 0.131 after. Variance decomposition shows that 94% of the reported effect’s variance originated from harness sampling noise, leaving an $sd \approx 0.18$ for any real difference. Every effect in the literature-relevant range is below this threshold. - **Seed replicates cannot catch a shared-stream confound**. Supplying the same rng stream to seed replicates handed every replicate the same decoy assignment, replicating the confound rather than the effect. - **Interface fidelity** resolves at much finer scales: the MDE for Tanimoto is 0.0072 (2.9% of its dynamic range). A correctly-sized molecule placed at the reference centroid artificially recovers 51% of

reference contacts and 31% of typed interactions, showing that absolute fingerprint scores are predominantly positional. - **Same-site filtering** is critical. Of 99 candidate true-binder ceiling pairs, 7 bound entirely different sites, which would have artificially deflated the ceiling and inflated any reported percentage gains. - **Sanitisation filtering** silently discarded ~2.5% of generated molecules inside geometric measurements. Direct ATOM parsing proved vastly superior to standard RDKit ([RDKit 2024](#)) sanitisation.

4 Discussion

We have demonstrated a sharp boundary in the capabilities of pretrained molecular-interaction foundation models. The model exhibits near-perfect within-system discrimination but fails to transfer this power cross-system. The mechanism underlying this failure suggests that optimising an abstract interface distance trades directly against physical pocket fit. When employed as a critic, the model successfully minimises its internal objective while degrading downstream molecular quality and docking performance.

Importantly, our detection limit analysis underscores the necessity of rigorous controls. Absolute metrics in SBDD are highly positional, and seed replicates often mask shared-stream confounds. Therefore, we recommend that evaluations of structure-based generative models must report a measured detection limit and a same-scale null, rather than relying solely on point estimates.

Data, scripts, and code availability

All source code, analysis scripts, and raw per-pair scores (in `results/specificity/` and `results/interface_fidelity/`) are openly available under the MIT License at the `ATOMICA-Diffusion-Antibiotic-design` repository. The pre-registration analysis plans (`results/*/ANALYSIS_PLAN.md`) were committed prior to results generation.

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Artificial intelligence disclosure

The core algorithms, software architecture, and numerical experiments were designed and directed by the author. Generative-AI coding assistants were used to help draft portions of the manuscript text; all quantitative results were produced by the author’s code, and all AI-assisted text was manually reviewed, verified

against the analytical results, and checked for errors before inclusion. The author takes full responsibility for the content of the manuscript.

Author contributions

E.A.V. designed the experiments, performed all numerical analyses, and wrote the manuscript.

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References

- Baillif, B. 2024. “Benchmarking Structure-Based Three-Dimensional Molecular Generative Models Using GenBench3D: Ligand Conformation Quality Matters.” *arXiv Preprint arXiv:2407.04424*, ahead of print. <https://doi.org/10.48550/arXiv.2407.04424>.
- Bouysset, C., and S. Fiorucci. 2021. “ProLIF: A Library to Encode Molecular Interactions as Fingerprints.” *Journal of Cheminformatics* 13 (72). <https://doi.org/10.1186/s13321-021-00548-6>.
- Buttenschoen, Martin, Garrett M. Morris, and Charlotte M. Deane. 2024. “PoseBusters: AI-Based Docking Methods Fail to Generate Physically Valid Poses or Generalise to Novel Sequences.” *Chemical Science* 15 (9): 3130–39. <https://doi.org/10.1039/D3SC04185A>.
- Fang, A., M. Desgagné, Z. Zhang, et al. 2025. “Learning Universal Representations of Intermolecular Interactions with ATOMICA.” *bioRxiv*, ahead of print. <https://doi.org/10.1101/2025.04.02.646906>.
- Francoeur, P. G., T. Masuda, J. Sunseri, et al. 2020. “Three-Dimensional Convolutional Neural Networks and a Cross-Docked Data Set for Structure-Based Drug Design.” *Journal of Chemical Information and Modeling* 60 (9): 4200–4215. <https://doi.org/10.1021/acs.jcim.0c00411>.
- Gao, Bowen, Minsi Ren, Yuyan Ni, et al. 2024. “Rethinking Specificity in SBDD: Leveraging Delta Score and Energy-Guided Diffusion.” *arXiv Preprint arXiv:2403.12987*, ahead of print. <https://doi.org/10.48550/arXiv.2403.12987>.
- Harris, Charles, Kieran Didi, Arian Jamasb, et al. 2023. “PoseCheck: Generative Models for 3D Structure-Based Drug Design Produce Unrealistic Poses.” *NeurIPS 2023 Generative AI and Biology (GenBio) Workshop*.

- 206 Koes, D. R., M. P. Baumgartner, and C. J. Camacho. 2013. “Lessons Learned in Empirical Scoring with
207 Smina from the CSAR 2011 Benchmarking Exercise.” *Journal of Chemical Information and Modeling*
208 53 (8): 1893–904. <https://doi.org/10.1021/ci300604z>.
- 209 Li, X. et al. 2025. “CBGBench: Fill in the Blank of Protein-Molecule Complex Binding Graph.” *International Conference on Learning Representations (ICLR)*.
210
- 211 Radoux, C. J., T. S. G. Olsson, W. R. Pitt, C. R. Groom, and T. L. Blundell. 2016. “Identifying Interactions
212 That Determine Fragment Binding at Protein Hotspots.” *Journal of Medicinal Chemistry* 59 (9): 4314–
213 25. <https://doi.org/10.1021/acs.jmedchem.5b01980>.
- 214 RDKit. 2024. *Open-Source Cheminformatics*. <https://www.rdkit.org>.
- 215 Schneuing, Arne, Charles Harris, Yuanqi Du, et al. 2024. “Structure-Based Drug Design with Equivariant
216 Diffusion Models.” *Nature Computational Science*, ahead of print. [https://doi.org/10.1038/s43588-024-](https://doi.org/10.1038/s43588-024-00737-x)
217 [00737-x](https://doi.org/10.1038/s43588-024-00737-x).
- 218 Seo, S., and W. Y. Kim. 2024. “PharmacoNet: Deep Learning-Guided Pharmacophore Modeling for
219 Ultra-Large-Scale Virtual Screening.” *Chemical Science* 15 (46): 19473–87. [https://doi.org/10.1039/](https://doi.org/10.1039/d4sc04854g)
220 [d4sc04854g](https://doi.org/10.1039/d4sc04854g).
- 221 Su, M., Q. Yang, Y. Du, et al. 2019. “Comparative Assessment of Scoring Functions: The CASF-2016
222 Update.” *Journal of Chemical Information and Modeling* 59 (2): 895–913. [https://doi.org/10.1021/acs.](https://doi.org/10.1021/acs.jcim.8b00545)
223 [jcim.8b00545](https://doi.org/10.1021/acs.jcim.8b00545).
- 224 Trott, O., and A. J. Olson. 2010. “AutoDock Vina: Improving the Speed and Accuracy of Docking with a
225 New Scoring Function, Efficient Optimization, and Multithreading.” *Journal of Computational Chem-*
226 *istry* 31 (2): 455–61. <https://doi.org/10.1002/jcc.21334>.